```python
#Problem 1: BUGGY CODE – Fix Using super()
class Person:

    # Constructor
    def __init__(self, name, id):
        self.name = name
        self.id = id

    # To display person details
    def Display(self):
        print(self.name, self.id)


class Emp(Person):

    def __init__(self, name, id):
        super().__init__(name, id)

    def Print(self):
        print("Emp class called")


Emp_details = Emp("Mayank", 103)

# Calling parent class function
Emp_details.Display()

# Calling child class function
Emp_details.Print()
```

```
Mayank 103
Emp class called
```

```python
#Problem 2: Anagram Check
def check_anagram(str1, str2):
    str1 = str1.replace(" ", "").lower()
    str2 = str2.replace(" ", "").lower()

    if sorted(str1) == sorted(str2):
        print("The strings are Anagrams.")
    else:
        print("The strings are not Anagrams.")


str1 = input("Enter first string: ")
str2 = input("Enabacter second string: ")

check_anagram(str1, str2)
```

```
Enter first string:  listen
Enter second string:  silent
```

```
The strings are Anagrams.
```

```
#Problem 3: Armstrong Number
num = int(input("Enter a number: "))

original_num = num
digits = len(str(num))
sum = 0

while num > 0:
    digit = num % 10
    sum = sum + digit ** digits
    num = num // 10

if sum == original_num:
    print("The number is an Armstrong number.")
else:
    print("The number is not an Armstrong number.")
```

```
Enter a number:  153

The number is an Armstrong number.
```

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#Problem 4: Classes and Objects
#Scenario: Car Dealership's Inventory Management System
#Requirements
#Vehicle class with maximum speed and mileage
#Default color = "white"
#Method to assign seating capacity
#Method to display all properties
#Create two vehicle objects:
#Max sclass Vehicle:
# Constructor
class Vehicle:

    # Constructor
    def __init__(self, max_speed, mileage):
        self.max_speed = max_speed
        self.mileage = mileage
        self.color = "white"
        self.seating_capacity = 0

    # Method to assign seating capacity
    def assign_seating_capacity(self, seating_capacity):
        self.seating_capacity = seating_capacity

    # Method to display all properties
    def display_properties(self):
        print("Maximum Speed:", self.max_speed, "kmph")
        print("Mileage:", self.mileage, "kmpl")
        print("Color:", self.color)
```

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        print("Seating Capacity:", self.seating_capacity)


# Creating first vehicle object
car1 = Vehicle(200, 20)
car1.assign_seating_capacity(5)

# Creating second vehicle object
car2 = Vehicle(180, 25)
car2.assign_seating_capacity(4)


# Displaying properties
print("Vehicle 1:")
car1.display_properties()

print("\nVehicle 2:")
car2.display_properties()
```

```
Vehicle 1:
Maximum Speed: 200 kmph
Mileage: 20 kmpl
Color: white
Seating Capacity: 5

Vehicle 2:
Maximum Speed: 180 kmph
Mileage: 25 kmpl
Color: white
Seating Capacity: 4
```